ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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GREEN HYDROGEN MARKET LANDSCAPE

Technology Assessment and Strategic Positioning Version 2.0 · February 2026 Non-Confidential

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EXECUTIVE SUMMARY

The global green hydrogen market is poised for exponential growth, driven by decarbonization imperatives, supportive government policies, and the falling costs of renewable energy. This document provides an analysis of the current market landscape, including key growth drivers, end-use sectors, and prevailing technology trends. It offers a competitive assessment of established electrolyzer technologies (Alkaline, PEM, SOEC, AEM) and benchmarks them against Tobe Energy's innovative membrane-free electrolysis system, highlighting significant advantages in capital expenditure (CAPEX), electrical efficiency, and material simplicity. Furthermore, a compelling use case for green hydrogen in powering AI data centers is presented, demonstrating the technology's potential to provide cost-effective, zero-emission energy storage and operational resilience for energy-intensive industries. Tobe Energy is positioned to capture significant market share by offering a next-generation solution that fundamentally alters the economics of green hydrogen production.

1. GREEN HYDROGEN MARKET OVERVIEW

1.1 Market Size & Growth Drivers

The green hydrogen sector is experiencing unprecedented momentum, with projections indicating a potential global capacity reaching hundreds of gigawatts by the mid-2030s. This expansion is underpinned by a confluence of factors transforming the energy landscape:

- **Robust National Policies and Strategies**: Governments worldwide (e.g., European Union via the Green Deal and REPowerEU, United States through the Inflation Reduction Act, and strategic initiatives in China,

Japan, Australia) are implementing ambitious hydrogen strategies and providing substantial funding. (See: IEA - Global Hydrogen Review 2023)

- **Corporate Decarbonization Commitments**: A growing number of corporations are adopting net-zero emission targets, driving demand for green hydrogen as a solution to decarbonize their operations.
- **Declining Renewable Energy Costs**: The continuously decreasing levelized cost of electricity (LCOE) from solar photovoltaic (PV) and wind power makes green hydrogen production increasingly cost-competitive.
- **Demand from Hard-to-Abate Sectors**: Industries such as steel manufacturing (via direct iron reduction), ammonia production (for fertilizers and future fuels), chemicals, and heavy-duty transportation (trucking, shipping, aviation) view green hydrogen as a critical decarbonization lever.

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1.2 Key End-Use Sectors

- **Industrial Feedstock & Process Heat**: Replacing gray hydrogen in ammonia production and refineries; enabling green steel and low-carbon chemicals
- **Transportation**: Fueling fuel cell electric vehicles (FCEVs), particularly buses, heavy-duty trucks, trains, and emerging maritime and aviation applications
- **Energy Storage & Grid Balancing**: Providing long-duration energy storage by converting surplus renewable electricity into hydrogen, which can then be re-electrified to support grid stability

1.3 Demand & Pricing Trends

A primary industry goal is achieving cost parity between green hydrogen and conventional fossil-derived (gray or blue) hydrogen. This is being accelerated by:

- Falling electrolyzer CAPEX, with targets moving towards <$500/kW for established technologies and even lower for next-generation systems
- Increasing electrolyzer efficiencies and operational lifetimes

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2. REGULATORY AND POLICY ENVIRONMENT

Governmental support is a crucial enabler for the nascent green hydrogen industry.

2.1 Government Incentives

Mechanisms such as production and investment tax credits (e.g., U.S. IRA's 45V credit), grants for research and deployment, and loan guarantees are stimulating electrolyzer manufacturing and project development.

2.2 Standards & Certification

The development of clear standards and certification schemes for "green" or "renewable" hydrogen (e.g., CertifHy in Europe, U.S. DOE's Clean Hydrogen Production Standard guidance) is essential for market transparency and investor confidence.

2.3 Oklahoma Context

The state of Oklahoma presents a favorable environment with:

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- Abundant and low-cost wind energy resources (35% of state electricity generation)
- Existing energy infrastructure and a skilled workforce
- A supportive business climate for energy sector investments
- Access to industrial clusters (fertilizer plants, refineries, chemical producers)

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3. ELECTROLYZER TECHNOLOGY LANDSCAPE

Several electrolyzer technologies are at different stages of maturity, each with distinct characteristics.

3.1 Established and Emerging Electrolyzer Technologies

TECHNOLOGY	CAPEX ($/KW)	ELECTRICAL EFFICIENCY	MATURITY (TRL)	NOTABLE CHARACTERISTICS
Alkaline (AEL)	$800–$900	~60-65%	High (9)	Well-established, robust, lower material costs, moderate efficiency
PEM	$1,100–$1,300	~55-65%	High (8-9)	Dynamic response, compact, but higher cost (PGM catalysts)
Solid Oxide (SOEC)	$4,000+	~70-80% (with heat)	Mid (6-7)	High efficiency (utilizes high-temp steam), complex, costly
Anion Exchange (AEM)	>$900	~55-60%	Mid (6-8)	Emerging; aims to combine AEL cost benefits with PEM flexibility

Note: CAPEX and efficiency figures are indicative and vary by manufacturer and scale; TRL = Technology Readiness Level

3.2 Tobe Energy's Membrane-Free Electrolysis System

METRIC	TOBE ENERGY ELECTROLYZER
CAPEX (Current Prototype)	**$446/kW**
Projected CAPEX (MW Scale)	**~$177/kW**
Electrical Efficiency (Stack, LHV)	**94.7%**
System Efficiency (HHV)	**>92% system efficiency (HHV)**
System Energy Consumption	**~42 kWh/kg H_2**
Stack Lifetime (Est.)	**~80,000+ hours**
Materials	Primarily 304 Stainless Steel

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3.3 Key Differentiators for Tobe Energy

1. **Breakthrough efficiency**: At **94.7% stack electrical efficiency** (LHV) and **>92% system efficiency** (HHV), Tobe's system significantly surpasses current industry standards, directly reducing the primary operational cost (electricity).
2. **Radically low CAPEX**: Projected MW-scale CAPEX of **~$177/kW** is substantially lower than any current commercial offering, transforming project economics.
3. **Durability and simplicity**: Utilization of common materials like 304 Stainless Steel, with no exotic or rare-earth elements, simplifies manufacturing, lowers maintenance (OPEX), and enhances supply chain resilience.
4. **Membrane-free design**: Eliminates the single largest cost and failure point in conventional electrolyzers, dramatically improving reliability and reducing costs.

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4. COMPETITIVE LANDSCAPE

The electrolyzer market features established players and emerging innovators.

4.1 Leading Players & Strategies

- **Alkaline Leaders** (e.g., Nel Hydrogen, Thyssenkrupp nucera): Leveraging decades of experience to focus on large-scale (gigawatt) deployments and achieving cost reductions through manufacturing volume and standardized designs.

- **PEM Innovators** (e.g., Plug Power, Cummins/Accelera): Targeting applications requiring flexible operation and rapid response (e.g., grid balancing, mobility) and offering integrated hydrogen solutions (electrolyzers, fuel cells, storage).
- **SOEC Pioneers** (e.g., Sunfire GmbH, Bloom Energy): Differentiating through high electrical efficiency, especially when integrated with industrial waste heat, though facing challenges in cost reduction and high-temperature material durability.
- **AEM Startups** (e.g., Enapter): Focusing on modular, scalable systems and aiming to combine the cost advantages of alkaline with the performance benefits of PEM, while addressing historical stability challenges.

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4.2 Positioning Tobe Energy

FEATURE	TOBE ENERGY	ALKALINE	PEM	SOEC	AEM
CAPEX	**Exceptional**	Moderate	Weak	Weak	Moderate
Electrical Efficiency	**Outstanding** (94.7% stack / >92% system)	Low (~62%)	Low (~61.5%)	High (~78%)	Low (~57%)
Stack Lifetime & Costs	**Excellent**	Good	Limited History	Moderate	Evolving
Scalability (Cost Curve)	**Strong**	Promising at Scale	Improving	Uncertain	Uncertain but Early
Commercial Readiness	**Mature** (Pilot+)	Mature	Mature	Emerging	Emerging

4.3 Strategic Insight

Tobe Energy is not offering an incremental improvement but a step-change in both production cost and electrical efficiency. This positions the company to potentially leapfrog incumbents by fundamentally altering the economic viability of green hydrogen projects.

5. MARKET OPPORTUNITIES & STRATEGIC CONSIDERATIONS FOR TOBE ENERGY

5.1 Industrial Decarbonization

Immediate opportunities exist to partner with entities in ammonia, refining, and steel sectors actively seeking cost-effective green hydrogen to meet decarbonization mandates. These industries represent the largest near-term addressable market for Tobe's technology.

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5.2 Leveraging Regional Advantages

Oklahoma and similar regions offer access to low-cost renewable electricity and a supportive environment for establishing initial production hubs. The state's wind resources (35% of electricity mix) and industrial infrastructure create an ideal launchpad for Tobe's commercialization.

5.3 Exploiting Early Mover Advantage

Tobe Energy's disruptive efficiency and projected cost metrics provide a powerful differentiating factor in a market eager for economic breakthroughs. The **$1.81/kg LCOH** target positions Tobe to undercut SMR costs without subsidies in many markets.

5.4 Focus on Scaling & Validation

Successful deployment of pilot and initial commercial-scale projects will be critical to validate CAPEX targets, long-term reliability, and operational performance at scale. The planned **Q4 2026** deployment timeline aligns with emerging market demand.

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6. USE CASE: GREEN HYDROGEN FOR COST-EFFICIENT, ZERO-EMISSION AI DATA CENTER POWER

6.1 The AI Energy Challenge

The exponential growth of Artificial Intelligence (AI) is creating an unprecedented surge in energy demand for data centers. This presents a unique market opportunity for innovative energy solutions that are both cost-effective and sustainable.

AI data centers require massive, reliable power. Simultaneously, there is intense pressure to reduce their carbon footprint. While nuclear, solar, and wind are primary zero-emission generation sources, the intermittency of renewables and the need for large-scale, cost-effective energy storage remain significant hurdles.

6.2 Nuclear Energy & Off-Peak Arbitrage

Nuclear power plants provide consistent baseload power but typically cannot ramp production up or down quickly to match fluctuating demand. Consequently, utilities often offer significantly discounted off-peak electricity rates (e.g., >60% cheaper than peak rates from utilities like Duke Energy). Capturing this low-cost off-peak power requires efficient, scalable energy storage.

6.3 Green Hydrogen Storage vs. Conventional Batteries

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- **Lithium-ion Batteries**: While effective for short-duration storage, batteries face challenges with long-duration needs, high capital costs for large-scale deployment, and reliance on critical minerals with volatile supply chains.
- **Tobe Energy's Hydrogen Solution**: The Tobe Electrolyzer can produce green hydrogen using low-cost off-peak electricity. This hydrogen can be stored efficiently and cost-effectively at scale (e.g., analysis suggests storing energy equivalent to what would require $30,000 of batteries could be achieved for less than $2,500 in hydrogen storage infrastructure, primarily driven by the electrolyzer cost). Hydrogen acts as an "energy battery" without the material constraints of lithium-ion.

6.4 Why Tobe Energy Enables This Now

Previous hydrogen production technologies were often inefficient and capital-intensive. Tobe Energy's membrane-free electrolysis, with its novel power electronics and simplified design, achieves significantly higher efficiency and lower CAPEX, making this energy arbitrage model economically viable.

6.5 Impact on AI Data Centers

- **Significant Cost Savings**: Produce and store green hydrogen using off-peak electricity (potentially ~60% cheaper), then convert it back to electricity via fuel cells during peak hours, drastically reducing overall energy expenditure.
- **Unmatched efficiency at low cost**: Tobe's system is projected to be up to 75% cheaper to build (CAPEX per kW of H_2 production capacity) and offers ~45% higher electrical efficiency compared to conventional electrolyzers, making it an affordable and sustainable energy source for AI's intensive demands.
- **Zero-Emission Resilience**: Pairing off-peak nuclear (or renewable) power with on-site green hydrogen production and fuel cell re-electrification creates a fully zero-emission power cycle, with water as the only byproduct. This enhances energy security, reduces carbon footprints, and alleviates grid stress.

6.6 Future-Proofing AI Expansion

Tobe's hydrogen solution enables AI data centers to:

1. Capitalize on low-cost off-peak electricity.

2. Minimize reliance on expensive, resource-constrained battery systems.
3. Maintain a sustainable, flexible, and scalable energy model aligned with long-term carbon reduction goals.

By integrating green hydrogen produced by Tobe Electrolyzers, AI data centers and other energy-intensive industries can meet escalating power demands cost-effectively and sustainably.

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7. CONCLUSION

The green hydrogen market is at an inflection point, demanding solutions that deliver both economic viability and high performance. Tobe Energy's membrane-free electrolysis technology — characterized by its **94.7% stack efficiency** (LHV), **>92% system efficiency** (HHV), significantly lower projected CAPEX (**~$177/kW**), and use of robust, common materials — is poised to disrupt the existing market landscape. This positions Tobe Energy as a key enabler for widespread green hydrogen adoption across industries.

The AI data center use case further exemplifies the transformative potential of Tobe's technology, offering a clear pathway to cost-effective, zero-emission energy storage and operational resilience. By overcoming the traditional cost and efficiency barriers, Tobe Energy is not just participating in the green hydrogen transition but is actively shaping its future, empowering new and existing markets to achieve their decarbonization goals sustainably and economically.